second term, which disappears when $N = Z$, in other words, when the number of protons equals the number of neutrons, and in which case the quantity of binding energy reaches the maximum. This means that from the point of view of energy, this is the most favourable situation. But this is not the case with the heavier atoms, where the last term produces a perceptible decrease of the binding energy. As regards the energy situation, it is more profitable that the second term should increase slightly, in consequence of N/Z slightly exceeding unity, in order that a very considerable diminution in the fourth term may thereby be achieved.

The binding energy is, in fact, a function of both N and Z, and we can represent it as such in a three-dimensional co-ordinate system, recording the number of protons on the horizontal axis running from left to right, the number of neutrons on the horizontal axis running from the front to the rear, and the binding energy on the vertical axis. Our equation will then produce a surface, for a particular energy value belongs to each pair of magnitudes (N, Z). The shape of the energy surface supplies all the information necessary for determining the stability of the nucleus. Figure 12 shows this surface, in the form of a map, with contour lines representing equal magnitudes of binding energy. The energy unit used is 0·001 a.m.u. Since the binding energies are negative, the surface lies below the plane of the drawing, like a valley dropping away from the lower left corner of the drawing toward the upper right corner (in other words, from the south-west to north-east, to keep the map analogy). The stable nuclei lie at the bottom of this valley, like houses along a gently winding street. Those nuclei which contain the same number of particles, always lie on straight lines, running from north-west to south-east at an angle of less than 45 deg. Among these nuclei, the one situated nearest to the bottom is always the most stable one. In so far as it is compatible with the conservation laws, one of the less stable nuclei can always change into a more stable one. The nuclei lying on the left side of the valley floor have too many neutrons, so that they must change by the emission of an electron. Those on the right side have too many protons, and would have to change by the emission of a positron. We find, however, that this very often does not happen, and there frequently exist two or three stable nuclei having the same $N + Z$ total. These nuclei are called *nuclear isobars*. This situation can be explained by certain finer details of the valley bottom only, not immediately evident from our equation based on very general assumptions. Actually, the valley bottom shows small convolutions and other fine details, on which actual measurements of the binding energies will supply information. We may say, nevertheless, that the stable nuclei will lie at the valley bottom or in its most immediate neighbourhood; we can also name the most stable of those nuclei which contain a certain given number of particles.